WEIZHEN WANG

School of Automation and Intelligent Sensing, Shanghai Jiao Tong University
Tel +8619821665689 Email: wangwz.im@gmail.com
Homepage Google Scholar

EDUCATION

Shanghai Jiao Tong University *Sept. 2023 - Present*
Ph.D. student in Automation Advisor: Prof. Xiaoming Duan

Zhejiang University *Aug. 2019 - Jun. 2023*
B.E. in Marine Engineering and Technology Rank: 1/82

RESEARCH INTERESTS

Reinforcement learning, robotic patrolling, Markov chains, Markov decision processes, motion planning.

AWARDS AND HONORS

- National Scholarships, Ph.D. Student, (top 0.2%), SJTU, 2025
- Provincial Outstanding Graduate Student, ZJU, 2023
- National Scholarships, Undergraduate Student, (top 0.2%), ZJU, 2020, 2021,2022
- First Prize in National College Student Mathematics Competition (Non-Mathematics Major Category), ZJU, 2020
- Second Prize in Zhejiang Province College Student Physics Innovation Competition, ZJU, 2020

PUBLICATIONS

Journal Articles

[1] **W. Wang**, X. Duan and J. He. Multi-Robot Stochastic Patrolling via Graph Partitioning. IEEE Transactions on Control of Network Systems, doi: 10.1109/TCNS.2024.3463502.

[2] **W. Wang**, J. He and X. Duan. On the Trade-Off Between Efficiency and Unpredictability in Stochastic Robotic Surveillance. IEEE Control Systems Letters, doi: 10.1109/LCSYS.2024.3515858.

[3] **W. Wang**, X. Duan and J. He. Growing Well-connected Networks for Entropy Rate Maximization. IEEE Transactions on Automatic Control, doi: 10.1109/TAC.2025.3575684.

[4] **W. Wang**, X. Zhou, J. He and X. Duan. Analysis of On-policy Policy Gradient Methods under the Distribution Mismatch. arXiv, doi: 10.48550/arXiv.2503.22244. [arXiv]

[5] X. Duan, **W. Wang** and R. Yan. Maximizing Markov Trajectory Entropy under Kemeny Constraints for Robotic Surveillance. IEEE Transactions on Automatic Control, doi: 10.1109/TAC.2026.3662306.

[6] **W. Wang**, Z. Wang, J. He, X. Guan and X. Duan. Minimizing Worst-Case Weighted Latency for Multi-Robot Persistent Monitoring: Theory and RL-Based Solutions. arXiv, doi: 10.48550/arXiv.2605.09633. [arXiv]

Conference Articles

[1] **W. Wang** and X. Duan. Joint Optimization of Defense Allocations and Surveillance Strategies against Random Intruders. Accepted by IFAC—PapersOnLine. [PDF]

TEACHING EXPERIENCE

Teaching Assistant SJTU

AU7036: Introduction to Multi-agent Systems Spring 2025

AU4606: Network Optimization Fall 2023, 2024, 2025, 2026

PROFESSIONAL ACTIVITIES

Presentations

- Oral Presentation at the IFAC August, 2026
  "Joint Optimization of Defense Allocations and Surveillance Strategies against Random Intruders"
  Busan, South Korea
- Oral Presentation at Chinese Conference on Swarm Intelligence and Cooperative Control November, 2025
  "Multi-Robot Stochastic Patrolling via Graph Partitioning" Shanghai, China
- Oral Presentation at the IEEE American Control Conference July, 2025
  "On the Trade-Off Between Efficiency and Unpredictability in Stochastic Robotic Surveillance."
  Denver, America
- Oral Presentation at Peking University October, 2024
  "Multi-Robot Stochastic Patrolling via Graph Partitioning" Beijing, China

Peer Reviewer

- IEEE Transactions on Control of Network Systems
- IEEE Network Magazine
- ACM Symposium On Applied Computing
- IEEE Control Systems Letters
- IEEE Conference on Decision and Control